2 June 1955

SUPERSEDING
NAVORD OS 1693C
29 July 1954

MILITARY SPECIFICATION

PRIMER, COMBINATION, MARK 15 MODS 1 AND 2

1. SCOPE

1.1 This specification governs the manufacture, assembly, loading, and preparation for delivery of Primer, Combination, Mk 15 Mods 1 and 2.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

LLL-B-631	-- Boxes; Fiber Corrugated (For Domestic Shipment).

BUREAU OF ORDNANCE

NAVORD OS 919	-- Priming Caps.
NAVORD OS 1666	-- Powder, Propellent; U. S. Navy Guns Ballistics Appendix, Navy.

STANDARDS

MILITARY

MIL-STD-105	-- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-- Marking for Shipment and Storage.
MIL-STD-352	-- Forty (40) Foot Drop Test for Use in Production of Fuzes.
MIL-STD-353	-- Transportation Vibration Test for Use in Production of Fuzes.

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BUREAU OF ORDNANCE

NAVORD OSTD 80 -- Sampling Procedures and Tables for Inspection by Variables.

DRAWINGS

BUREAU OF ORDNANCE

SK 41232 -- Lock Primer Mk 15 Mod 1, List of Drawings.
SK 109599 -- Lock Primer Mk 15 Mod 2, List of Drawings.
SK 167001 -- Primer Container for Primer Mk 15 Mods 1 and 2, and Mk 16 Mod 0, List of Drawings.
And all documents listed on the above Lists of Drawings.

PUBLICATIONS

BUREAU OF ORDNANCE

NAVORD OD 5823 -- Sensitivity Test of Primers and Detonators.
OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the Procuring Activity or as directed by the Contracting Officer.)

2.2 Other publications. -- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 -- Transportation, Interstate Commerce Commission, Explosives, and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950).")

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AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM Designation: B 154-51 -- Mercurous Nitrate Test for Copper and Copper Alloys.

(Copies of ASTM Standards may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania).

3. REQUIREMENTS

3.1 General.

3.1.1 Primers, Combination, Mk 15 Mod 1 covered by this specification shall be manufactured in accordance with Bureau of Ordnance List of Drawings SK No. 41232 and all documents listed thereon.

3.1.2 Primers, Combination, Mk 15 Mod 2 covered by this specification shall be manufactured in accordance with Bureau of Ordnance List of Drawings SK No. 109599 and all documents listed thereon.

3.1.3 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Bridge wire subassembly continuity. -- Before assembling the bridge wire subassembly, it shall be thoroughly cleaned to remove solder flux and contamination. After cleaning, the bridge wire shall be flashed by application of an electric current to assure continuity of both legs of the bridge wire. The current shall be regulated to produce a visible red glow of the bridge. An EMF of approximately 1.5 volts ac or dc may be used with a suitable series resistor to adjust the generator characteristics to the primer resistance. Flashing is a necessary manufacturing operation and shall be conducted on each assembly. If the bridge wire of any assembly fails to glow, the assembly shall be screened from acceptable material in the process. The discarded assemblies may be reworked by replacement of the bridge.

3.3 Bridge wire resistance. -- There shall be no short or open circuits. The plunger insulator shall be firmly held by the stock crimp and shall not be broken or displaced. The resistance of the bridge shall be between 0.55 and 0.70 ohms when tested as specified in 4.5.1.

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3.4 Percussion functioning. -- The primer shall function when the plunger is actuated by the impact of a 16.34 ounce steel ball falling from a height of 10.0 inches. Functioning shall not result in blowback, split stock, or detachment of metallic components from the primer. The primer shall not function when the plunger is impacted by a 16.34 ounce steel ball falling from a height of 3.0 inches. The test provisions of 4.5.2 shall be applicable in ascertaining conformance.

3.5 Primer sensitivity. -- The sensitivity of the primer shall be ascertained as prescribed in 4.5.3. The sensitivity values shall be used for data card record and process control. Deviations in sensitivity values shall not directly cause rejection of the primers, but may be reason for checking materials and processes for conformance with drawing and specification requirements.

3.6 Metallic seals, mercury cracking. -- Metallic seals shall show no splits or cracks when subjected to the mercurous nitrate test of 4.5.4.

3.7 Electrical functioning. -- Each lot of primers shall pass the static electric firing test specified in 4.5.5.

3.8 Vibration. -- The primers shall be safe and operable after vibration in accordance with the provisions of Military Standard MIL-STD-353 and 4.5.6.

3.9 Forty (40) foot drop test. -- The primers shall withstand the drop test as prescribed in Military Standard MIL-STD-352 and 4.5.7.

3.10 Ballistic performance. -- The primers shall give proper ignition and there shall be no misfires or hangfires. Projectile ejection shall occur within .100 second after application of the firing energy. The primers shall provide ignition which permits meeting the velocity uniformity requirements of NAVORD OS 1666. Function shall not produce splitting or distortion which prevents ready extraction from the primer lock. The ballistic performance test of 4.5.8 shall be applicable in ascertaining conformance, except that the alternative simulated ignition test of 4.5.8.1 shall be applicable when specially authorized by the Government.

3.11 Primer caps. -- Primer caps used in assembly shall contain none of the defects specified in OS 919 and shall be within the specified sensitivity range. The primer cap shall not be deformed in any way during insertion or in final assembly.

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3.11.1 After insertion of the primer cap into the plunger cup, the assembly shall be gaged to insure that the primer cap has been properly seated.

3.12 Moisture content.

3.12.1 Black powder. -- The moisture content of the black powder used in the metallic seal and ignition cup mixture shall not exceed 0.70 percent.

3.12.2 Gun cotton and pyrocellulose. -- The moisture content of the gun cotton and pyrocellulose shall not exceed 0.50 percent

3.13 Primer weighing. -- The Contractor shall check weigh all completed primers to insure presence of black powder charge.

3.14 Final assembly. -- Final assembly of the primer by the loading activity shall be in accordance with the applicable drawing. All component parts shall be thoroughly cleaned and dried before assembly and operations shall be conducted so as to prevent inclusion of moisture, oil, and foreign matter in the primer. The method of loading and assembling shall be uniform for each lot. After the paper disc is placed on top of the metallic seal charge, it shall be given a coating of shellac and the shellac allowed to dry completely before assembling the metallic seal in the primer stock. Assemble the metallic seal with the primer stock. Seal the end of the primer between the primer stock and metallic seal with shellac and allow to dry before final packaging. Where parts involved are of plastic or fiber, such parts after final assembly and machining shall be kept in airtight containers until ready for use in final primer assembly.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of Standard MIL-STD-105.

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4.1.1 Lot. -- As applied to the Government inspection of units of product, the term "Lot" shall mean "Inspection Lot", i.e., a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in " Inspection Lots" shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of three hundred (300) complete Primers, Combination, Mk 15 of one Mod shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These primers shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Production lot.-- For procurement purposes the production lot shall consist of 5,000 primers of one Mod plus those required for all test purposes. Unless otherwise stated in the contract, order or requisition, all units for test purposes shall be at the expense of the Contractor. The production lot should not be confused with the lot for inspection purposes defined in 4.1.1.

4.4 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government inspection shall be in accordance with the provisions of General Procedures for Government Inspection and Contactor's Inspection shall be in accordanee with Inspection by Contractor, as contained in OP 400.

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4.5 Tests.

4.5.1 Bridge wire resistance test. -- Completed primers shall be tested for bridge wire resistance using a test circuit which limits the bridge current to 50 millamperes maximum. Primers with bridge wire resistance greater than 0.70 ohm or less than 0.55 ohm shall be rejected.

4.5.2 Percussion functioning test. -- Fit Test Set Mk 135 Mod 0 with a firing chamber and plunger actuator. Mounting of the primer shall simulate support against the forces of explosion equivalent to service operation. Adjust the scale and align the test set, as provided in NAVORD OD 5823, to deliver impacts coincident with the axis of the plunger. Conduct one (1) test for functioning at a drop height of 10.0 inches. Defectives for this test include failure to fire, hangfire which is obvious to operator without instrumentation, blowback, split stock, and detached metallic components. Conduct a second test for not functioning at a drop height of 3.0 inches. Primers which function shall be considered defective. Sampling and disposition of the lot shall be on the basis of each individual lot in accordance with Military Standard MIL-STD-105, Inspection Level I, Multiple Sampling Plan, AQL 0.25 percent defective.

4.5.3 Primer sensitivity test. -- The test shall be conducted on the basis of 50 primers from each production lot. Test the primers in Test Set Mk 135 Mod 0, NAVORD OD 5823, under the mounting provisions of 4.5.2. The staircase method (Bruceton) as prescribed in NAVORD OD 5823 shall be applicable. Use a 16.34 ounce drop ball and 0.50 inch drop increments starting at a suggested height of 6.0 inches for the test. Compute the 50 percent firing height ($\bar{h}$) and the standard deviation (s), and record the values on the data card.

4.5.4 Metallic seals, mercurous nitrate test. -- Metallic seals shall be subjected to the test prescribed in ASTM B 154-51, Mercurous Nitrate Test for Copper and Copper Alloys. Seals which do not split or crack shall be acceptable. Disposition of a maximum lot of 25,000 seals shall be in accordance with Military Standard MIL-STD-105, Table VI-1, Single Sampling Plan, AQL 1.0 percent defective. Metallic seals which have been subjected to this test shall be scrapped.

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4.5.5 Electrical functioning test. -- The test shall be conducted in a safety chamber which simulates support against the forces of explosion equivalent to service operation. Half of the sample primers shall be subjected to a d-c potential difference of two (2) volts, and half of the sample primers shall be subjected to a d-c potential difference of twenty (20) volts. The primer stock and plunger shall serve as terminals. It is important that firm contact on clean bright surfaces be obtained to avoid high contact resistance. Disposition of a lot of primers shall be in accordance with NAVORD OSTD 80, Test Level I, AQL 2.5 percent defective, using an upper specification limit of twenty (20) milliseconds. In the disposition of a lot of more than 8,000 primers, one (1) only misfire or firing time above the upper specification limit may be selectively assigned the value of twenty (20) milliseconds. The firing times for the two specified voltages shall be combined into a single set of values for computing the acceptance values under this test.

4.5.6 Vibration test. -- Rigidly mount twenty-four (24) primers from each production lot to the vibration table. The sample orientations for the three-phase test shall be axis-vertical, metallic seal-down; axis-vertical, metallic seal-up; and axis-horizontal. Subject the primers to the vibration schedule of Military Standard MIL-STD-353. The lot shall be rejected if one (1) of the primers explodes or is rendered unsafe to handle and use for service purposes. The lot shall be rejected if more than one (1) primer fails to meet the bridge resistance requirements of 3.3. Divide the sample into three (3) groups of eight (8) primers each. Subject one (1) group of eight (8) primers to the electrical functioning test of 4.5.5. Subject the two (2) remaining groups to the percussion functioning test of 4.5.2. One (1) group shall be subjected to the ten (10) inch drop test and one (1) to the three (3) inch drop test. The incidence of more than one (1) functioning defect shall cause rejection of the lot. The incidence of one (1) defect shall cause a retest on the basis of a sample of forty-eight (48) primers with rejection of the lot on obtaining more than one (1) defective.

4.5.7 Forty (40) foot drop test. -- Nine (9) primers from each production lot shall be subjected to the drop test prescribed in Military Standard MIL-STD-352. Each primer shall be assembled in fixture which will insure rigid mounting of the primer. The fixture and not the primer shall receive the drop impact. Three (3) of the primers shall be dropped to strike axis-vertical, metallic seal-down; three (3) axis-vertical, metallic seal-up; and three (3) axis-horizontal. The lot shall be acceptable for this test if none of the primers fire or are judged to be unsafe to handle or transport.

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4.5.8 Ballistic performance test. -- Primers as listed below shall be tested on the basis of each production lot for performance in a bag gun for which the primer is a service item. Blind-loaded or target projectiles shall be used as components of the round. The bag charges for service pressure firing shall be from a lot of known acceptable characteristics. The listed firing table, method of firing and associated defects shall be applicable in disposition of the lot.

Class A Defect	Class B Defect
Detached metallic component	Hangfire (ejection time greater than .100 second)
Blow back	Velocity dispersion outside limits of NAVORD OS 1666
Misfire	
Distortion involving extraction difficulty	

Class A Defects Only at Proof Pressure

Sample Size	Method of Firing	Acceptance No.	Reject No.
5	3 Percussion 2 Electrical	0	1

Service Pressure

Cumulative Sample Size	Method of Firing	Class A Defect Accept No.	Class A Defect Reject No.	Class B Defect Accept No.	Class B Defect Reject No.
15	7 Percussion 8 Electrical	0	1	0	3
45	22 Percussion 23 Electrical	0	1	2	3

4.5.8.1 Simulated ignition test. -- The simulated ignition test as herein prescribed may be used in lieu of the ballistic performance test of 4.5.8 when authorized by the Government. Sample primers shall be fired in standard firing locks in a bag gun for which the primer is a service item and using dummy charges with live ignition ends. Disposition shall be on the basis of each production lot in accordance with that part of the firing table in 4.5.8 titled "Service Pressure". Performance shall be judged and acceptance made only on the basis of the "Class A Defects" listed in the firing table.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Inert parts. -- Unless otherwise specified by the contract or order, all inert parts or components shall be packaged in compliance with the requirements of Specification LLL-B-631. Each container shall incorporate corrugated or solid fiberboard dividers or collars suitable for preventing damage to the threaded areas in handling and transportation. The gross weight of the container and contents shall not exceed 70 pounds.

5.1.2 Complete primers. -- Seventy-five (75) Primers Mk 15 type, shall be packaged in accordance with Bureau of Ordnance List of Drawings SK 167001.

5.2 Packing. -- Six (6) unit containers shall be packed in accordance with Bureau of Ordnance List of Drawings SK 167001.

5.3 Marking. -- In addition to any special marking required by the contract or order, all containers shall be marked in accordance with Standard MIL-STD-129 and Code of Federal Regulations 49 CFR 71-78.

5.4 Data cards. -- Data cards five inches by eight inches, printed as shown in Figure 1, shall be properly filled out for each primer lot. One card shall be placed in the top of each container, five in each shipping box, two shall accompany ballistic samples, five shall be forwarded to the Procuring Agency, and twenty shall go to the Ammunition Depot receiving the primers.

6. NOTES

6.1 Intended use. -- The Primers, Combination, Mk 15 Mods 1 and 2 are used to ignite the propellant charge in all bag guns.

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FIGURE 1

AMMUNITION DATA CARD

Type, Mark & Mod_____ Lot No._____

Project Order No._____ Lot Size_____

Loaded by_____ Date_____

Percussion Sensitivity h_____ s_____

Inspected by_____ Commanding Officer_____

COMPONENTS	MANUFACTURER	Lot No.	Year Mfd.
Inert parts set			
Black powder FFG			
Primer cap			
Black powder FFFG			

Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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